Olist E-Commerce Performance & Operations Analysis

(2016–2018)

Project Overview

Business Problem:

You are a Data Analyst hired by Olist, a Brazilian e-commerce marketplace. The business is growing but leadership doesn't know *where* revenue is coming from, *why* customer satisfaction is low in certain regions, and *which* sellers and categories to prioritize. Your job is to find answers using 2 years of transaction data and give actionable recommendations.

Central Business Question:

Which product categories, states, and seller segments are driving Olist's revenue and satisfaction – and what operational changes will improve growth and retention over the next 2 quarters?

Executive Summery

Olist experienced strong marketplace growth between October 2016 and August 2018, generating approximately

$13.73M in revenue from

94K orders and serving

91K customers.

The analysis shows that revenue is heavily concentrated in a small number of states, product categories, and sellers. While overall delivery performance is strong (**90.5% on-time delivery rate**), operational inefficiencies in specific regions continue to impact customer satisfaction and retention.

Executive Summery

Three key findings emerged:

1. **Revenue concentration risk**

    São Paulo (SP) is the dominant revenue driver.

    A limited number of product categories generate a disproportionately large share of marketplace revenue.

2. **Delivery performance drives customer satisfaction**

    Late deliveries account for approximately **9.5% of orders**
    Review scores decline significantly when delivery expectations are missed.

3. **Customer retention is weak**

    Repeat purchase rate is only **2.76%**.
    Most customers make a single purchase and never return.

Executive Summery

These findings suggest that Olist's next phase of growth should focus on:

- Improving logistics performance in underperforming states.
- Increasing customer retention through loyalty and re-engagement initiatives.
- Prioritizing high-performing sellers and categories while addressing seller quality issues.

Dataset Overview

Time Period: October 2016 – August 2018

MarketPlace Performance:

Total Revenue:
$13.73M

Total Orders:
94K

Total Customer:
91K

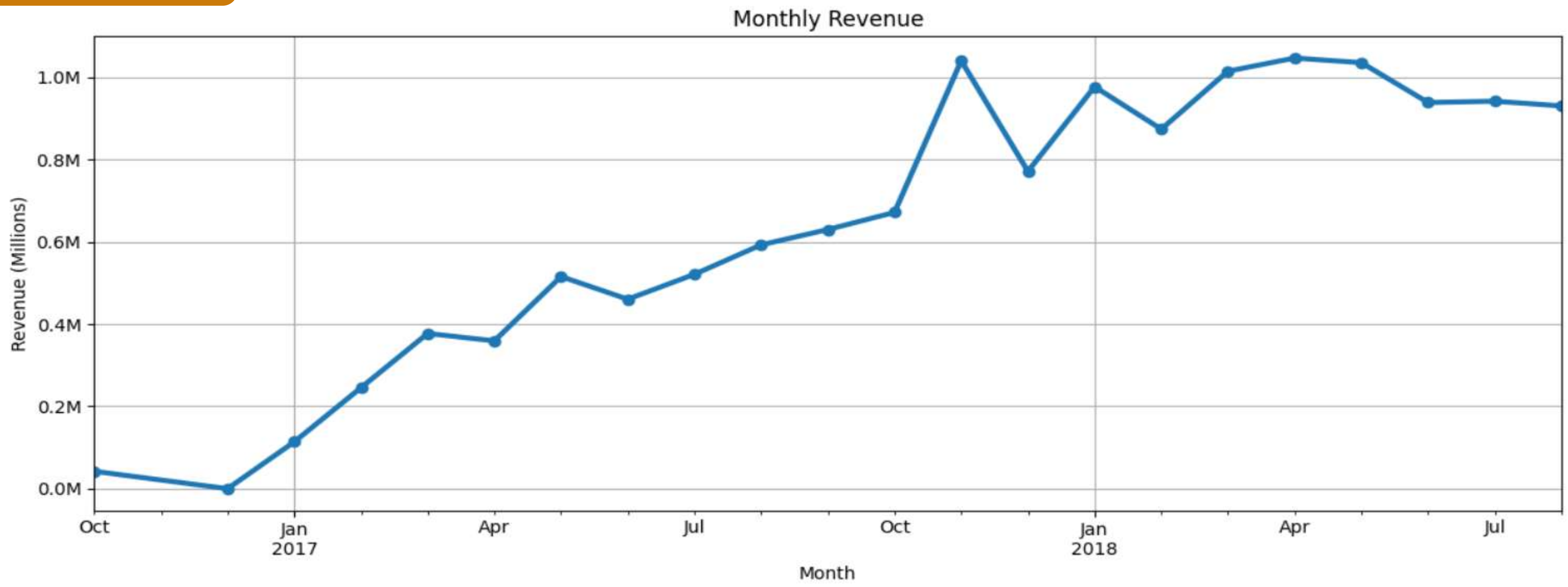
Avg Review Score:
3.0/5.0

Successful Deliveries:
90.5%

Avg Delivery Time:
12 Days

Revenue Analysis

Revenue Trend:



Strong upward trajectory during 2017.

Revenue exceeded $1M per month by late 2017.

Revenue stabilized at a high level throughout 2018.

Revenue Analysis

Top Revenue Product Categories:

1. **Health & Beauty (~$1.35M)**
2. **Watches & Gifts (~$1.24M)**
3. **Bed, Bath & Table (~$1.12M)**
4. **Sports & Leisure (~$1.04M)**
5. **Computers & Accessories (~$0.91M)**

These categories represent the marketplace's primary revenue engines

Business implication:
Marketing spend, inventory expansion, and seller acquisition efforts should prioritize these categories because they have already demonstrated product-market fit.

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6. **Furniture décor (~$0.75M)**
7. **Housewares (~$0.68M)**
8. **Cool Stuff (~$0.67M)**
9. **Auto (~$0.63M)**
10. **Toys (~$0.53M)**

Recommendation:
These categories generate significantly lover revenue ,
Conduct category profitability analysis before increasing investment in lower-performing segments.

Revenue Analysis

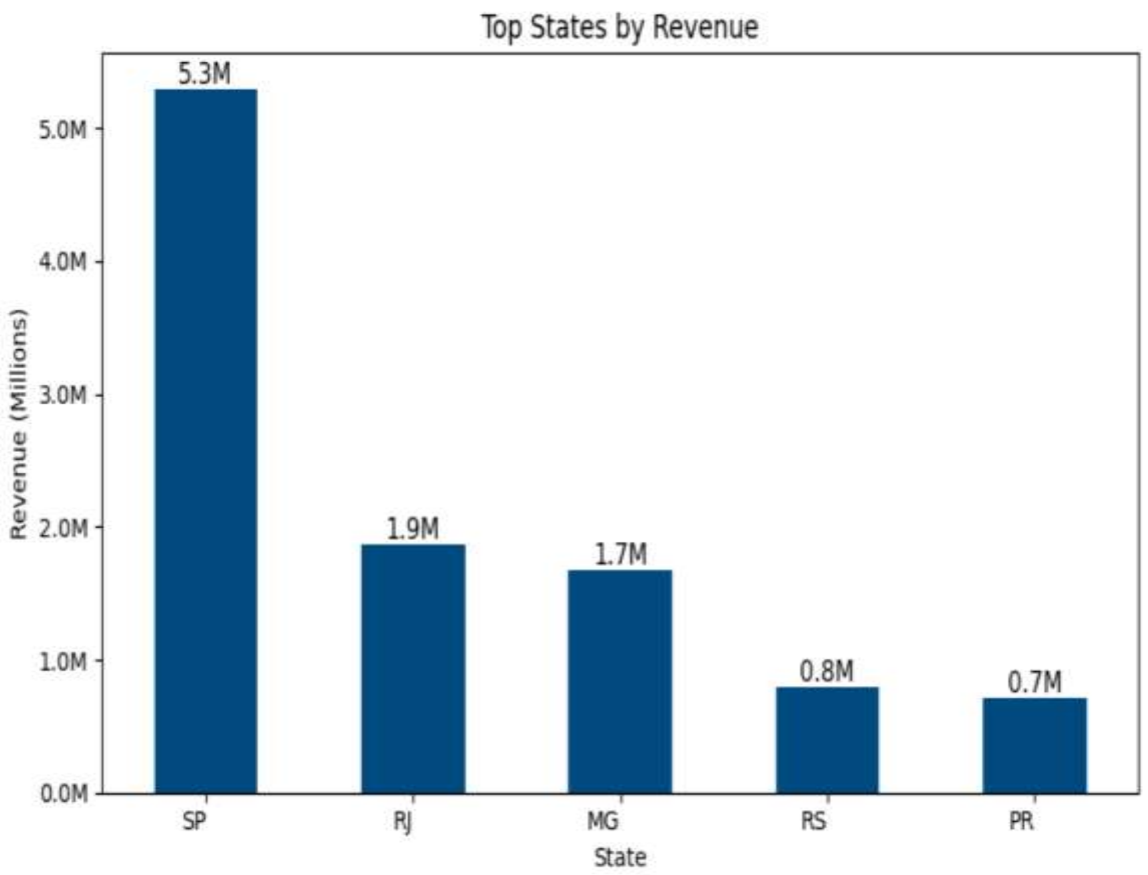
Top Revenue States:

Revenue is highly concentrated geographically.
Top contributors:

1. **São Paulo (SP)** **(~$5.3M)**
2. **Rio de Janeiro (RJ)** **(~$1.9M)**
3. **Minas Gerais (MG)** **(~$1.7M)**

64% Total Revenue

São Paulo alone contributes substantially more revenue than any other state.



Business implication:

Olist is highly dependent on SP, creating concentration risk.
Develop growth campaigns to diversify revenue sources targeting:

1. **Paraná (PR)**
2. **Santa Catarina (SC)**
3. **Distrito Federal (DF)**
4. **Goiás (GO)**

Revenue Analysis

Why Revenue is very high in certain states?

From state revenue analysis it tis clear that: **Total Orders ↑ = Total Revenue ↑**

For Every State if:

Avg Delivery Time ↓

Avg Fright Value ↓

No Of Orders ↑ → **Total Revenue ↑**

Avg Delivery time and **Avg Fright Value** depends upon distance between seller and customer.

Example:
Almost **60% Sellers** are from **São Paulo (SP)** **(1.73K)**
therefore delivery time and fright value is less for SP -> More Customer from SP -> More Revenue From SP

Delivery Overview

Key KPIs:

- **Freight Revenue: $1.97M**
- **Average Freight Cost: $20.90**
- **Canceled Orders: 428**
- **Late Deliveries: 9.5%**

→

Overall performance is strong, with nearly 91% of deliveries arriving on time.

Delivered orders account for approximately 98% of completed fulfillment activity.
This demonstrates operational stability at scale.

For Every State If:

Late Delivery Rate ↑ → ↓ Avg Customer Review

Avg Delivery Time ↑ → ↑ Avg Fright value

- If Customer recives late delivery they give less review score
- As the delivery time increases for a order the fright value also increases

Customer Behaviour

Key KPIs:

- **Total Customer 91K**
- **Total Repeat Customer: 3K**
- **Average Customer Value: $154.16**
- **Repeat Buy Rate: 2.76%**

Interpretation:
Customer acquisition is healthy, but customer retention is weak.

Business impact:

Acquiring customers is substantially more expensive than retaining existing ones.

Recommendation:

Implement:

- Loyalty programs
- Personalized recommendations
- Post-purchase engagement campaigns
- Targeted email marketing
- to increase repeat purchasing.

Customer Review

Most reviews are positive:

- **Score 5: 58.6%**
- **Score 4: 19.5%**
- **Score 1: 10.3%**
- **Remaining scores account for smaller proportions.**

Despite this, the overall average review score remains relatively modest because negative experiences are concentrated among specific customer segments.

Satisfaction Drivers:

The strongest driver of dissatisfaction appears to be delivery performance.

Evidence:

- Most Late deliveries correlate with lower review scores. **Score 1: 40%**
- Regions with longer delivery times show lower customer satisfaction.

Recommendation:

Improving logistics performance is likely to generate a larger satisfaction improvement than expanding product assortment.

Customer Segment

Lost Customers **~58%** (Largest Customer Segment)

Characteristics:

- Long time since last purchase
- mostly bought once
- low engagement and revenue

New Customers **~37%**

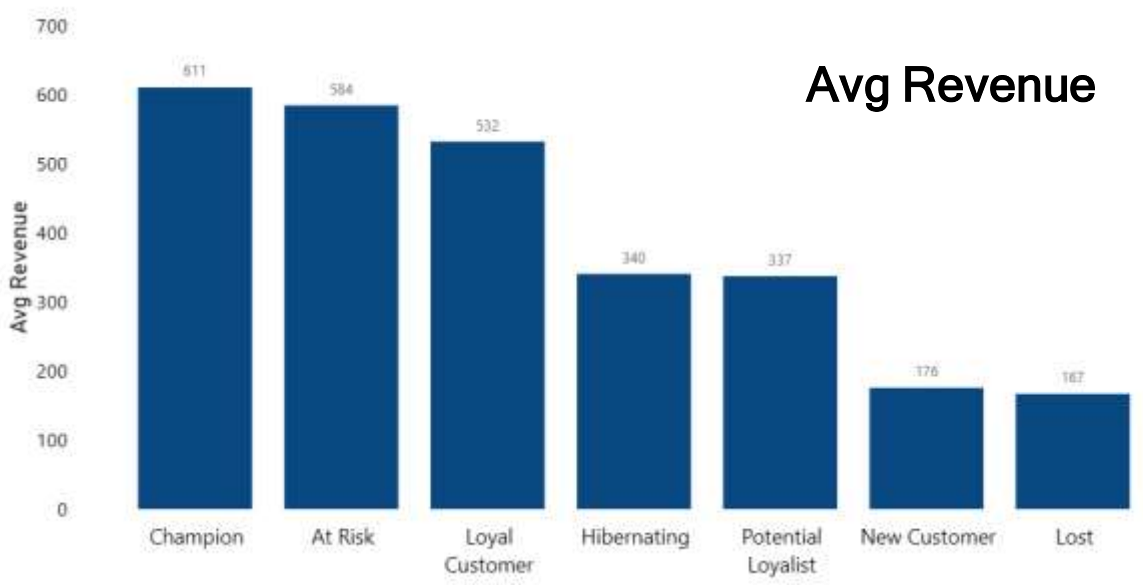
Characteristics:

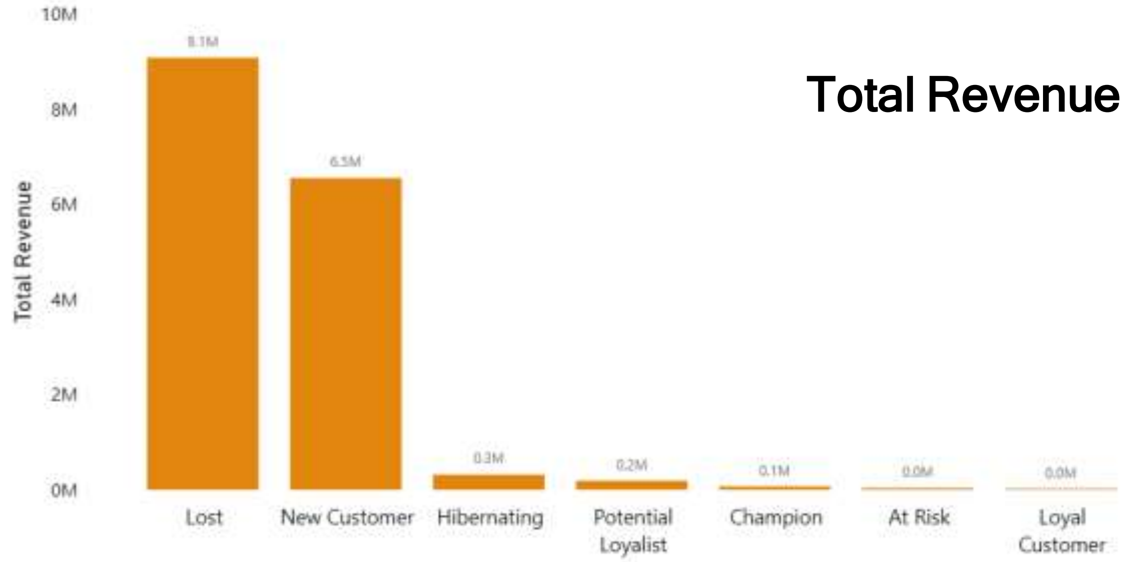
- Long time since last purchase
- mostly bought once
- low engagement

Champion Customers **< 1%**

Characteristics:

- High purchase frequency & recency
- Highest individual spent





Strategic Implications

Retain Champions

Provide:

- Exclusive promotions
- VIP rewards
- Early product access

Convert New Customers

Focus on:

- Second-order incentives
- Loyalty enrollment
- Follow-up engagement

Reactivate Lost Customers

Launch:

- Win-back campaigns
- Discount offers
- Personalized product recommendations

Seller Analysis

Key KPIs:

- **Total Sellers: 2907**
- **Avg Revenue Per Seller: $4.87K**
- **Avg Orders Per Seller: 34**
- **Avg Seller Rating: 4.13/5**

Revenue Concentration Among Sellers:

A small number of sellers generate disproportionately high revenue.
This indicates a marketplace power-law structure where top sellers drive a large share of GMV.

Business implication:

Protecting and supporting high-performing sellers is critical.

Seller Analysis

Seller Quality Assessment:

Analysis identified risk sellers characterized by:

- High revenue contribution
- Low customer review scores

These sellers create significant business risk because they affect many customers.

Recommendation:

Implement:

- Seller scorecards
- Performance monitoring
- Mandatory improvement plans

for underperforming high-volume sellers

Seller Review:

Late Delivery Rate ↑ → ↓ Avg Customer Review

- Seller from **AM** has highest late delivery rate **~67%**
- Sellers from **AM** has Lowest avg review **~2.3/5**

Seller State Performance:

Seller revenue is concentrated in:

- **São Paulo** **~65%** (Of Total Revenue)
- **Paraná**
- **Minas Gerais**

These states form the operational backbone of the marketplace.

Recommendation

Priority 1: Improve Delivery Performance

Actions:

- Identify late-delivery hotspots.
- Rebalance logistics partners.
- Introduce SLA monitoring.

Expected outcome:

- Higher customer satisfaction.
- Fewer negative reviews.

Priority 2: Increase Customer Retention

Actions:

- Loyalty program.
- Win-back campaigns.
- Personalized recommendations.

Expected outcome:

- Higher repeat purchase rate.
- Lower customer acquisition costs.

Priority 3: Double Down On High-Revenue Category

Focus:

- Health & Beauty
- Watches & Gifts
- Bed, Bath & Table
- Sports & Leisure.

Expected outcome:

- Faster revenue growth.
- Higher marketing efficiency.

Conclusions

The analysis shows that Olist's growth is being driven primarily by a concentrated group of product categories, sellers, and states—**particularly São Paulo** and the **Health & Beauty** segment.

While marketplace revenue growth remains strong, the greatest opportunity for the next two quarters is not customer acquisition **but operational optimization** and **customer retention**.

By improving **delivery reliability**, **strengthening seller quality controls**, and **increasing repeat purchases,** Olist can simultaneously improve customer satisfaction, retention, and sustainable revenue growth.